

DATA WAREHOUSING

- Data warehousing is the process of collecting, integrating, storing and managing data from multiple sources in a central repository.
- It enables organizations to organize large volumes of current and historical data for efficient querying, analysis and reporting.
- The main goal of data warehousing is to support decision-making by providing clean, consistent and timely access to data. It ensures fast data retrieval even when working with massive datasets.

NEED FOR DATA WAREHOUSING

- **Handling Large Data Volumes:** Traditional databases store limited data (MBs to GBs), while data warehouses are built to handle huge datasets (up to TBs), making it easier to store and analyze long-term historical data.
- **Enhanced Analytics:** Databases handle transactions; data warehouses are optimized for complex analysis and historical insights.
- **Centralized Data Storage:** A data warehouse combines data from multiple sources, giving a single, unified view for better decision-making.
- **Trend Analysis:** By storing historical data, a data warehouse allows businesses to analyze trends over time, enabling them to make strategic decisions based on past performance and predict future outcomes.
- **Business Intelligence Support:** Data warehouses work with BI tools to give quick access to insights, helping in data-driven decisions and improving efficiency.

COMPONENTS

- **Data Sources:** These are the various operational systems, databases and external data feeds that provide raw data to be stored in the warehouse.
- **ETL (Extract, Transform, Load) Process:** The ETL process is responsible for extracting data from different sources, transforming it into a suitable format and loading it into the data warehouse.
- **Data Warehouse Database:** This is the central repository where cleaned and transformed data is stored. It is typically organized in a multidimensional format for efficient querying and reporting.
- **Metadata:** Metadata describes the structure, source and usage of data within the warehouse, making it easier for users and systems to understand and work with the data.

COMPONENTS

- **Data Marts:** These are smaller, more focused data repositories derived from the data warehouse, designed to meet the needs of specific business departments or functions.
- **OLAP (Online Analytical Processing) Tools:** OLAP tools allow users to analyze data in multiple dimensions, providing deeper insights and supporting complex analytical queries.
- **End-User Access Tools:** These are reporting and analysis tools, such as dashboards or Business Intelligence (BI) tools, that enable business users to query the data warehouse and generate reports.

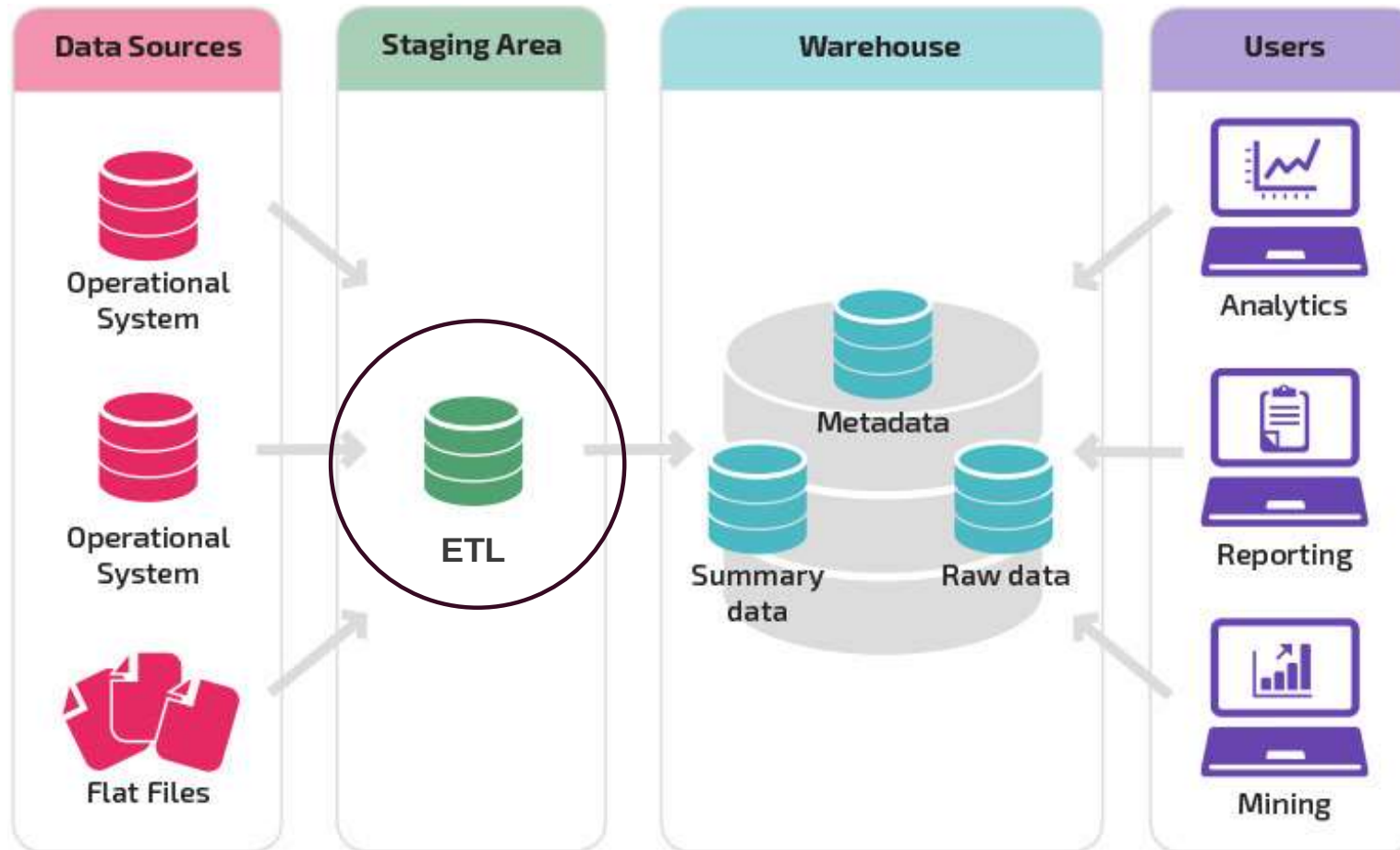
REAL WORLD EXAMPLE OF DATA WAREHOUSING

- Data Warehousing can be applied anywhere where we have a huge amount of data and we want to see statistical results that help in decision making.
- **1. E-commerce**
- **Data Gathering:** Orders, returns, payments, user clicks, delivery updates.
- **Schema:** Combines source data into a structured star schema for analysis.
- **Cleansing:** Standardizes customer names, locations and product categories.
- **Updates:** Near real-time or scheduled loads for fresh insights.
- **Summarization:** Bestsellers by category, regional demand trends, logistics performance.

BANKING

- **Data Gathering:** ATM transactions, online banking, credit card usage, loan records.
- **Schema:** Integrates data from core banking, CRM and fraud detection systems.
- **Cleansing:** Fixes inconsistencies in account info, transaction logs and addresses.
- **Updates:** Transaction data is batched and uploaded nightly.
- **Summarization:** Daily cash flow reports, high-risk account flags and customer profitability analysis.

Data Warehouse Architecture

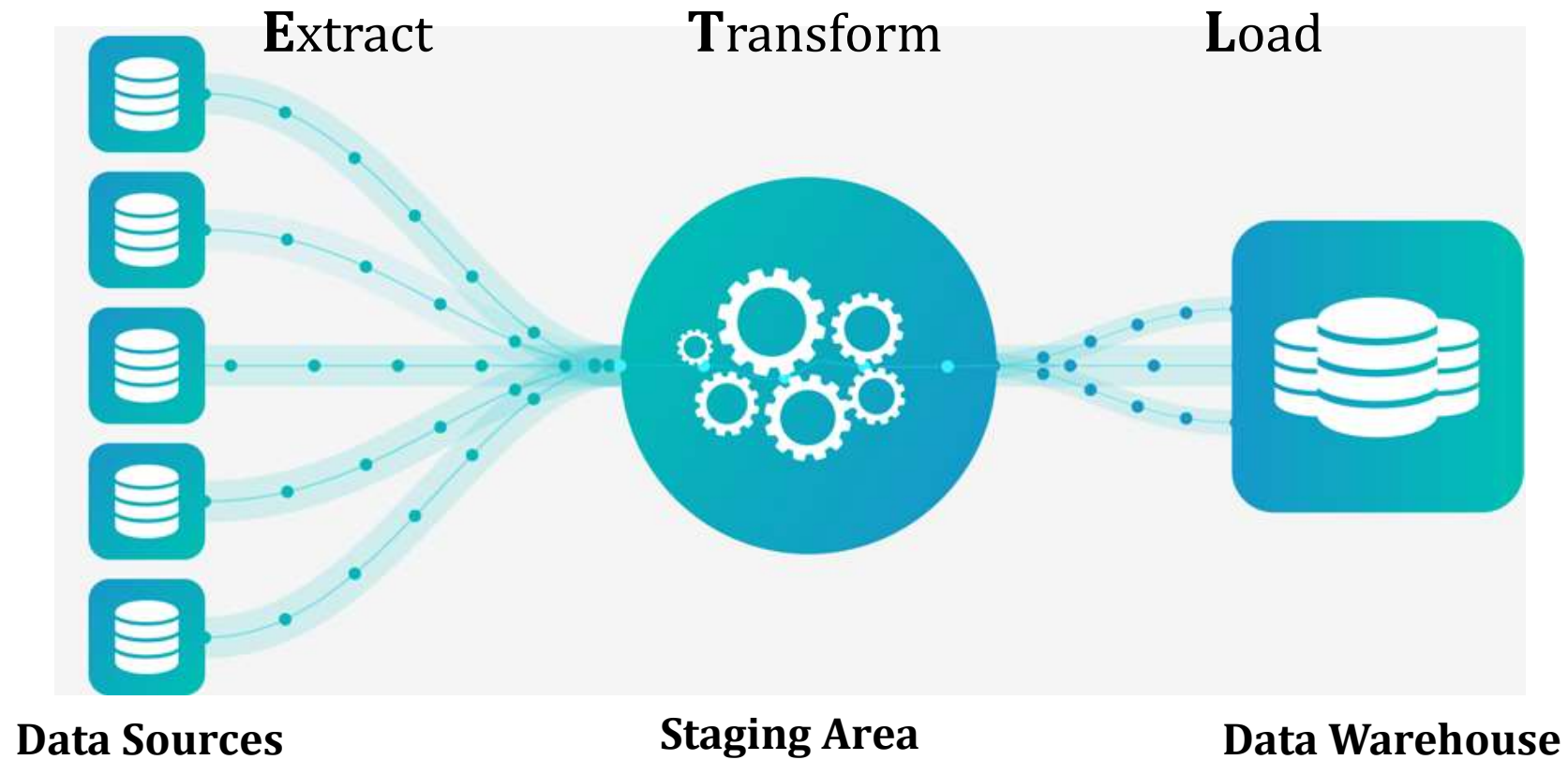




ETL Process



What is ETL?



What is ETL?

- **Extraction, Transform and Load**
- ***Data Extraction***
 - Extraction of data from the source systems.
- ***Data Transformation***
 - Includes all the functions and procedures for changing the source data into the exact formats and structures appropriate for storage in the data warehouse database
- ***Loading Data***
 - Physically moves the data into the data warehouse repository.

Steps in ETL

1. Determine all the target data needed in the data warehouse.
 2. Determine all the data sources, both internal and external.
 3. Prepare data mapping for target data elements from sources.
 4. Establish comprehensive data extraction rules.
 5. Determine data transformation and cleansing rules.
 6. Plan for aggregate tables.
 7. Organize data staging area and test tools.
 8. Write procedures for all data loads.
 9. ETL for dimension tables.
 10. ETL for fact tables.
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ETL Key Factors

- Complexity of data extraction and transformation
 - Reason: the tremendous **diversity** of the source systems.
 - Need to pay special attention to the various sources and begin with a complete inventory of the source systems.
 - The difficulties encountered in the data transformation function also relate to the **heterogeneity** of the source systems.
- Data Loading Functions
 - Find the proper time to schedule full refreshes.
 - Determine the best method to capture the ongoing changes from each source system
 - Execute the capture without impacting the source systems.
 - Schedule the incremental loads without impacting use of the data warehouse by the users



Data Extraction



Data Extraction

- Data extraction for a data warehouse
 - you have to extract data from many different sources
 - you have to extract data on the changes for ongoing incremental loads as well as for a one-time initial full load.
- Two factors increase the complexity of data extraction for a data warehouse.
- What are the Data Extraction issues??
 1. Source identification
 2. Method of extraction
 3. extraction frequency
 4. Time window
 5. Job sequencing
 6. Exception handling

Data Extraction

- Source Identification
 - It encompasses the identification of all the proper data sources
 - It include examination and verification that the identified sources will provide the necessary value to the data warehouse

Data Extraction Techniques

- Data in Operational Systems
 - Operational data in the source system falls into two broad categories.
 - The type of data extraction technique you have to use depends on the nature of each of these two categories.
 - **Current Value**
 - The stored value of an attribute represents the value of the attribute at this moment of time.
 - The values are transient or transitory.
 - There is no way to predict how long the present value will stay or when it will get changed next.
 - Data extraction for preserving the history of the changes in the data warehouse gets quite involved for this category of data.

Data Extraction Techniques

- Data in Operational Systems

- **Periodic Status**

- The value of the attribute is preserved as the status every time a change occurs.
 - At each of these points in time, the status value is stored with reference to the time when the new value became effective.
 - The history of the changes is preserved in the source systems themselves.
 - Therefore, data extraction for the purpose of keeping history in the data warehouse is relatively easier.

Data Extraction Techniques

- Initial load
- Two major types of data extractions from the source operational systems:
 - “As is” (static) Data Extraction
 - Data Extraction of Revisions

Data Extraction Techniques

- ***“As is” (static) data extraction***
 - The capture of data at a given point in time.
 - Taking snapshot of relevant source data at certain point in time
 - Primarily used for initial load of data warehouse
 - This data capture would include each status or event at each point in time as available in the source operational systems.

Data Extraction Techniques

- *Data of revision*
 - Also called as **incremental data capture**
 - Not strictly incremental data but the revisions since the last time data was captured
 - If the source data is transient, the capture of the revisions is not easy.
 - For periodic status data or periodic event data, extracts the statuses and events that have been recorded since the last data extraction.
 - Incremental data capture may be **immediate or deferred**.

Data Extraction Techniques

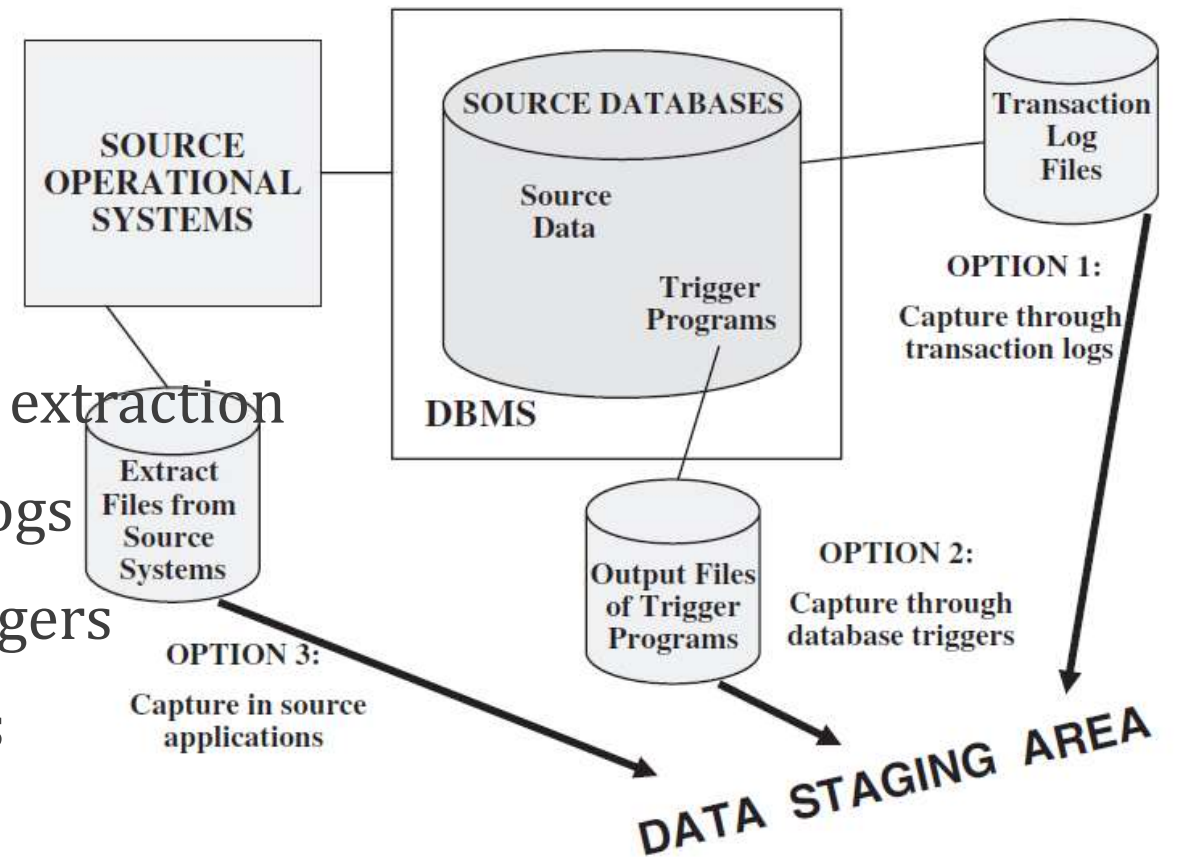
- *Data of revision*

- Immediate Data Extraction

- The data extraction is real-time.

- Three options for immediate data extraction

1. Capture through transaction logs
2. Capture through database triggers
3. Capture in source applications



Data Extraction Techniques

- *Data of revision*
 - Immediate Data Extraction: **Capture through transaction logs**
 - Uses the transaction logs of the DBMSs maintained for recovery from possible failures.
 - As each transaction adds, updates, or deletes a row from a database table, the DBMS immediately writes entries on the log file.
 - This data extraction technique reads the transaction log and selects all the committed transactions.
 - Make sure that all transactions are extracted before the log file gets refreshed
 - If source system data is on indexed and other flat files, this option will not work for these cases.

Data Extraction Techniques

- *Data of revision*

- Immediate Data Extraction: **Capture through Database Triggers**

- Create trigger programs for all events for which you need data to be captured.
 - The output of the trigger programs is written to a separate file that will be used to extract data for the data warehouse.
 - Data capture through database triggers occurs right at the source and is therefore quite reliable.
 - You can capture both before and after images.
 - Building and maintaining trigger programs puts an additional burden on the development effort.
 - Applicable only for source data in databases.

Data Extraction Techniques

- *Data of revision*

- Immediate Data Extraction: **Capture through Source Applications**

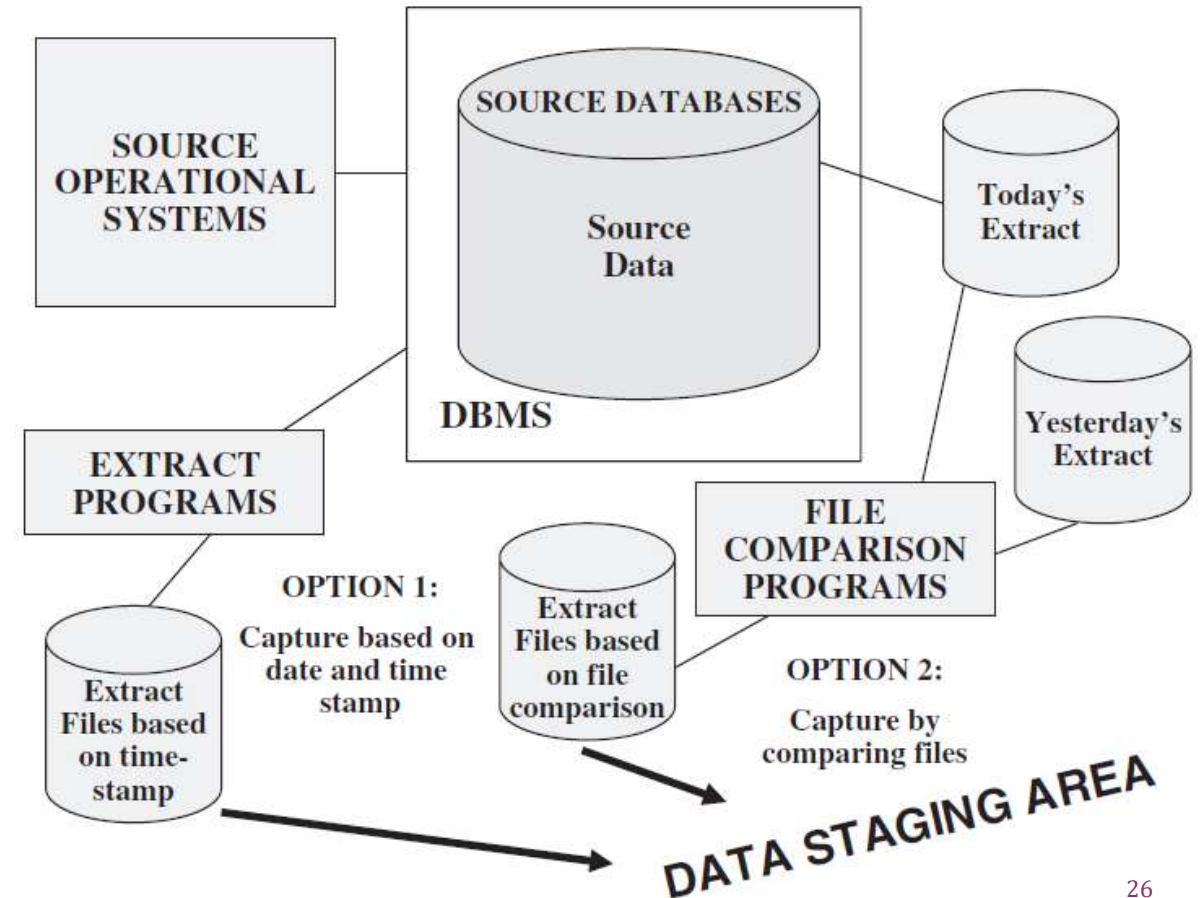
- Also referred to as application assisted data capture.
 - The source application is made to assist in the data capture for the data warehouse.
 - Revise the programs to write all adds, updates, and deletes to the source files and database tables.
 - Other extract programs can use the separate file containing the changes to the source data.
 - Applicable for all types of source data irrespective of whether it is in databases, indexed files, or other flat files.
 - May degrade the performance of the source applications because of the additional processing needed to capture the changes on separate files.

Data Extraction Techniques

- **Data of revision**

- Deferred Data Extraction:

- Do not capture the changes in real time.
 - The capture happens later.
 - Two options
 1. Capture based on date and time stamp
 2. Capture by comparing files



Data Extraction Techniques

- *Data of revision*

- Deferred Data Extraction: **Capture based on date and time stamp**

- Every time a source record is created or updated it may be marked with a stamp showing the date and time.
 - The time stamp provides the basis for selecting records for data extraction.
 - This technique works well if the number of revised records is small.
 - Data capture based on date and time stamp can work for any type of source file.
 - It captures the latest state of the source data.
 - Any intermediary states between two data extraction runs are lost.
 - Deletion of source records presents a special problem.
 - Mark the source record for delete, do the extraction run, and then physically delete the record

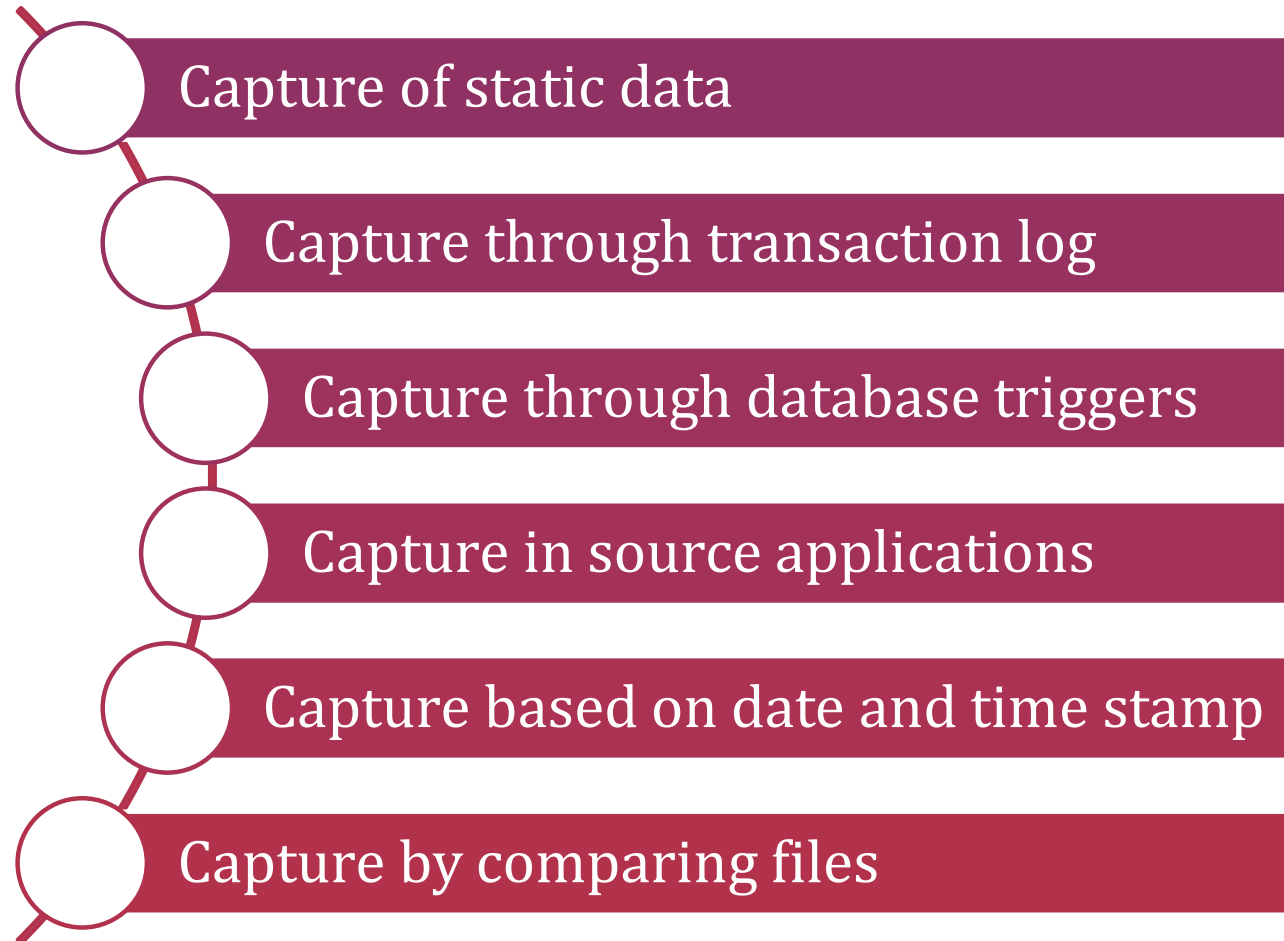
Data Extraction Techniques

- *Data of revision*

- Deferred Data Extraction: **Capture by Comparing Files**

- Also called the snapshot differential technique,
 - Do a full file comparison between today's copy of the data and yesterday's copy.
 - Compare the record keys to find the inserts and deletes.
 - Capture any changes between the two copies.
 - Requires to keep the prior copies of all the relevant source data.
 - Though simple and straightforward, comparison of full rows in a large file can be very inefficient.

Data Extraction Techniques



Data Transformation



Data Transformation

- The extracted data must be made usable in the data warehouse.
- We have to enrich and improve the quality of the data before it can be usable in the data warehouse.
- Basic Tasks in Data Transformation
 - **Selection**
 - Select either whole records or parts of several records from the source systems.
 - **Splitting / Joining**
 - Splitting the selected parts even further during data transformation.
 - Joining of parts selected from many source systems is more widespread in the data warehouse environment.

Data Transformation

- Basic Tasks in Data Transformation
 - **Conversion**
 - Includes fundamental conversions of single fields for two primary reasons
 - To standardize among the data extractions from disparate source systems,
 - To make the fields usable and understandable to the users.
 - **Summarization**
 - Summarizing data at different level of granularities
 - **Enrichment**
 - The rearrangement and simplification of individual fields to make them more useful for the data warehouse environment

Data Transformation

- Major Transformation Types / Transformation Functions
 - **Format Revisions**
 - Includes changes to the data types and lengths of individual fields.
 - It is wise to standardize and change the data type to text to provide values meaningful to the users.
 - **Decoding of fields**
 - You are bound to have the same data items described by a plethora of field values.
 - E.g. codes used for recording gender information
 - You need to decode all such cryptic codes and change these into values that make sense to the users.

Data Transformation

- Major Transformation Types / Transformation Functions
 - **Calculated and Derived values:**
 - e.g The sales system contains sales amounts, sales units, and operating cost estimates by product.
 - You will have to calculate the total cost and the profit margin before data can be stored in the data warehouse
 - **Splitting of single field**
 - By splitting the fields
 - You may improve the operating performance by indexing on individual components.
 - Users may need to perform analysis by using individual components
 - E.g address of an employee or customers

Data Transformation

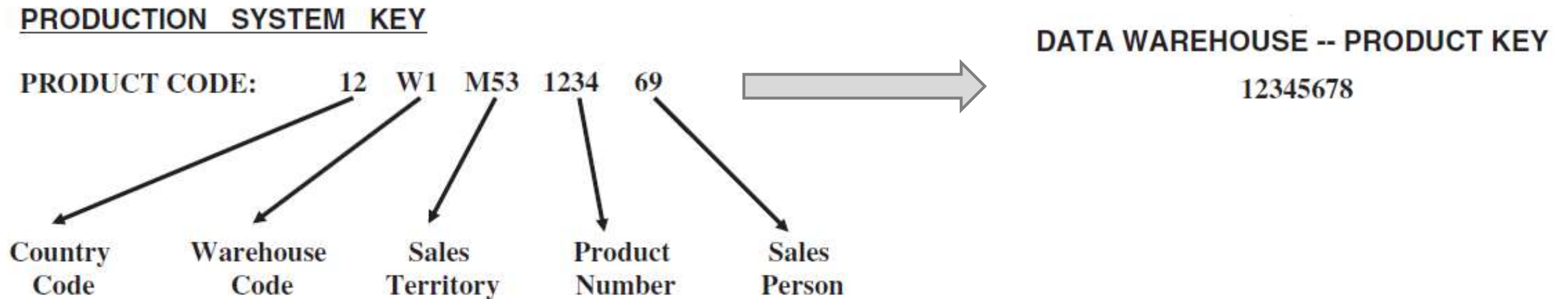
- Major Transformation Types / Transformation Functions
 - **Merging of Information:**
 - Merging of information denotes the combination of the different fields from different sources into a single entity.
 - **Character set conversion**
 - The conversion of character sets to an agreed standard character set for textual data in the data warehouse.
 - **Conversion of Units of Measurements**
 - You may have to convert the metrics so that the numbers are all in one standard unit of measurement.
 - E.g. currency

Data Transformation

- Major Transformation Types / Transformation Functions
 - **Date/Time Conversion:**
 - Representation of date and time in standard formats.
 - **Summarization:**
 - The creation of summaries to be loaded in the data warehouse instead of loading the most granular level of data.
 - **Deduplication**
 - Keeping a single record for one customer and link all the duplicates in the source systems to this single record.

Data Transformation

- Major Transformation Types / Transformation Functions
 - **Key Restructuring**
 - Transform primary keys of extracted records into generic keys generated by the system itself.



Data Transformation

- **Data Integration and Consolidation**

- The real challenge of ETL functions is the *pulling together of all the source data from many disparate, dissimilar source systems*.
- Integrating the data involves combining all the relevant operational data into coherent data structures to be made ready for loading into the data warehouse.
- Standardize the names and data representations and resolve discrepancies in the ways in which the same data is represented in different source systems.
- Data integration and consolidation is a type of preprocessing before any data transformation

Data Transformation

- **Data Integration and Consolidation**

- *Entity Identification Problem*

- e. g. Three different legacy applications have three different customer files supporting those systems.
 - The same customer on each of the files may have a unique identification number which may not be the same across the three systems.
 - In this case, you do not know which of the customer records relate to the same customer.
 - In the data warehouse you need to keep a single record for each customer.
 - Match up the activities of the single customer from the various source systems with the single record to be loaded to the data warehouse.

Data Transformation

- **Data Integration and Consolidation**

- *Entity Identification Problem*

- Users need to be involved in reviewing the exceptions to the automated procedures.
 - Solution is in two phases.
 - In the first phase, all records, irrespective of whether they are duplicates or not, are assigned unique identifiers.
 - The second phase consists of integrating the duplicates periodically through automatic algorithms and manual verification.

Data Transformation

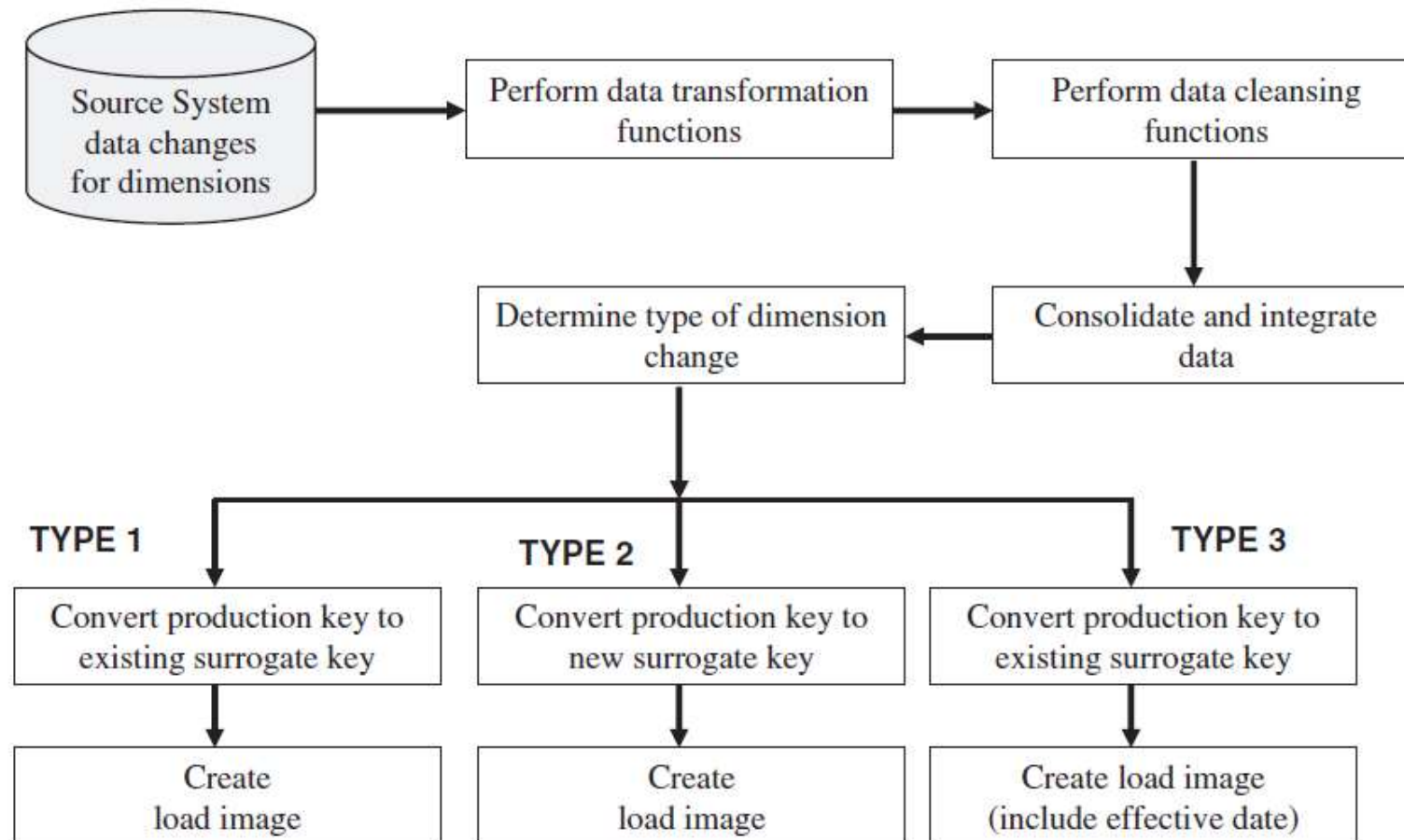
- **Data Integration and Consolidation**

- *Multiple Sources Problem*

- This problem results from a single data element having more than one source.
 - E.g. unit cost of products is available from two systems. From which system should you get the cost for storing in the data warehouse?
 - Assigning a higher priority to one of the two sources and pick up the product unit cost from that source.
 - Select from either of the files based on the last update date.
 - Determining of the appropriate source which depends on other related fields.

Data Transformation

■ Transformation for Dimension Attributes



Data Transformation

- **How to Implement Transformation?**

- The types of data transformation are by far more difficult and challenging.
- Data transformation tools can be expensive.

- ***Using Transformation Tools***

- Improves efficiency and accuracy.
- You just have to specify the parameters, the data definitions, and the rules to the transformation tool.
- When you specify the transformation parameters and rules, these are stored as metadata by the tool.
- When changes occur to transformation functions, the metadata for the transformations get automatically adjusted by the tool.

Data Transformation

- **How to Implement Transformation?**

- *Using Manual Techniques*

- Manual techniques are adequate for smaller data warehouses.
 - Manually coded programs and scripts perform every data transformation.
 - This method involves elaborate coding and testing.
 - Unlike automated tools, the manual method is more likely to be prone to errors.
 - In-house programs have to be designed differently if you need to store and use metadata.
 - Every time changes occur to transformation rules, the metadata has to be maintained.
 - This puts an additional burden on the maintenance of manually coded transformation programs

Data Loading

Data Loading

- Transformation functions end as soon as load images are created.
- Create load images to correspond to the target files to be loaded in the data warehouse database.
- Moving data into the data warehouse repository can be done in following ways
 - ***Initial load***
 - Populating all the data warehouse tables for the very first time
 - ***Incremental Load***
 - Applying ongoing changes as necessary in a periodic manner.
 - ***Full Refresh***
 - Completely erasing the contents of one or more tables and reloading with fresh data
 - Initial load is a refresh of all the tables

Data Loading

- During the loads, the data warehouse has to be **offline**.
- Need to find a **window of time** when the loads may be scheduled without affecting your data warehouse users.
- **Divide up the whole load process** into smaller chunks and populating a few files at a time
 - You may be able to run the smaller loads in parallel.
 - You might be able to keep some parts of the data warehouse up and running while loading the other parts.
- Provide procedures to handle the load images that do not load.
- Have a plan for quality assurance of the loaded records.

Data Loading

- Having staging area and data warehouse database on same server will save efforts.
- If they are on different servers, consider the options carefully and select one wisely.
 - The Web, FTP, and database links are a few of the options.
 - You have to consider the necessary bandwidth needed and also the impact of the transmissions on the network.
- *How can we apply data?*
 - By writing **special load programs**.
 - **Load utilities** that come with the DBMSs provide a fast method for loading.

Data Loading

- ***Techniques and Processes for Applying Data***
 - Three types of application of data to the data warehouse: *initial load, incremental load, and full refresh*
 - **For the initial load**
 - Extract the data from the various source systems,
 - Integrate and transform the data,
 - Then create load images for loading the data into the respective dimension table.
 - **For an incremental load**
 - Collect the changes to the product data for those product records that have changed in the source systems since the previous extract,
 - Run the changes through the integration and transformation process,
 - Create output records to be applied to the product dimension table.
 - **A full refresh** is similar to the initial load

Data Loading

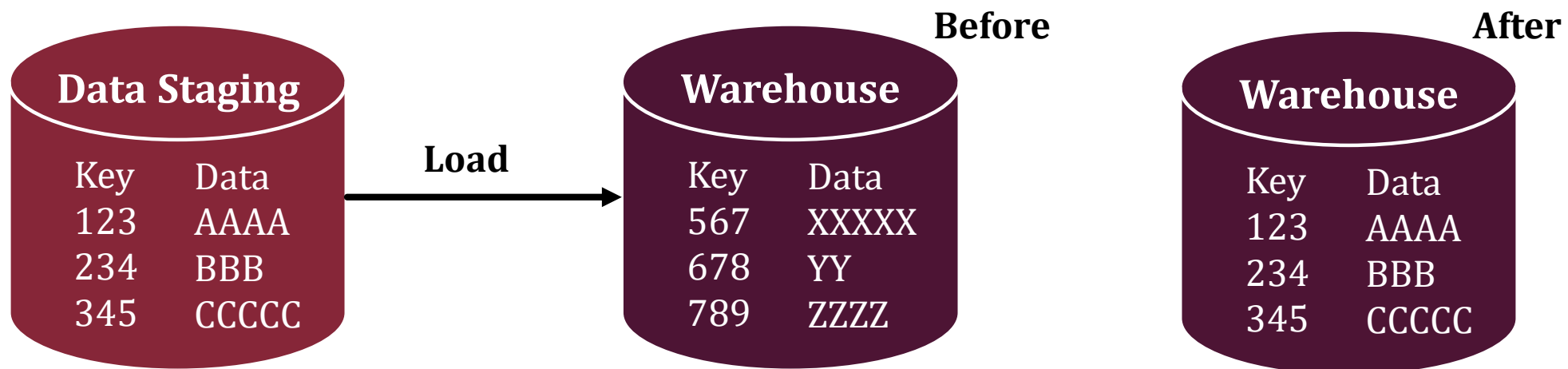
- *Techniques and Processes for Applying Data*
 - We need to create a file of data to be applied to the product dimension table in the data warehouse.
 - Data may be applied in the following four different modes:
 1. Load
 2. Append
 3. Destructive Merge
 4. Constructive Merge

Data Loading

■ *Modes for Applying Data*

1. Load

- If the target table to be loaded already exists and data exists in the table, the load process wipes out the existing data and applies the data from the incoming file.
- If the table is already empty before loading, the load process simply applies the data from the incoming file.



Data Loading

- *Modes for Applying Data*

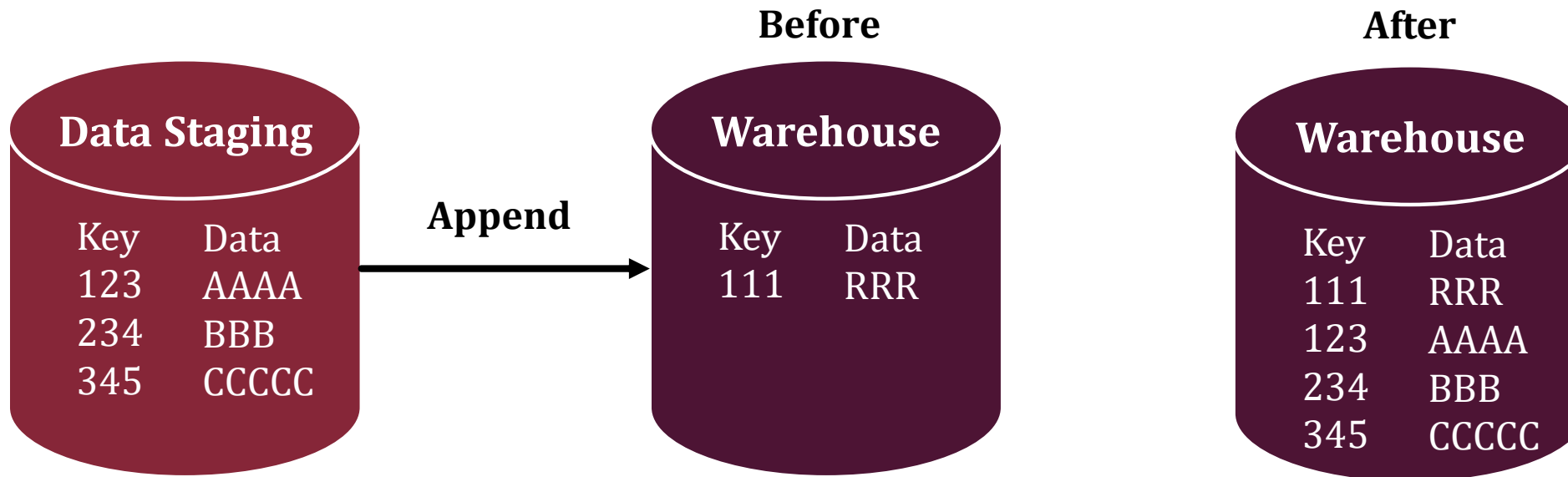
- 2. **Append**

- If data already exists in the table, the append process unconditionally adds the incoming data, preserving the existing data in the target table.
 - When an incoming record is a duplicate of an already existing record, you may define how to handle an incoming duplicate.
 - The incoming record may be allowed to be added as a duplicate.
 - the incoming duplicate record may be rejected during the append process.

Data Loading

- *Modes for Applying Data*

- 2. Append

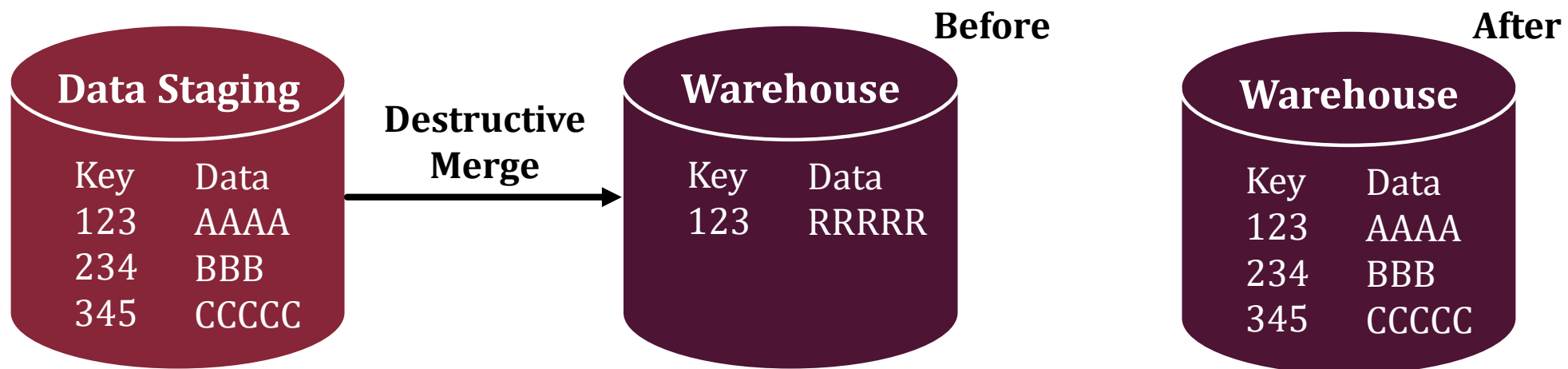


Data Loading

- *Modes for Applying Data*

- 3. **Destructive Merge**

- If the primary key of an incoming record matches with the key of an existing record, update the matching target record.
 - If the incoming record is a new record without a match with any existing record, add the incoming record to the target table.

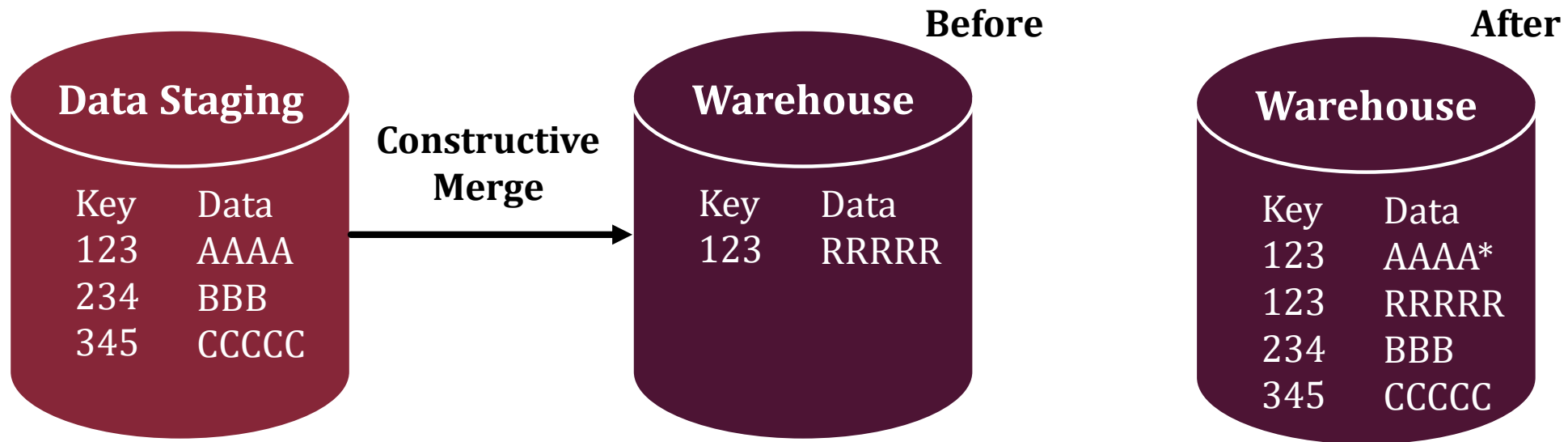


Data Loading

- *Modes for Applying Data*

- 4. Constructive Merge

- If the primary key of an incoming record matches with the key of an existing record, leave the existing record, add the incoming record, and mark the added record as superseding the old record.



Data Loading

- How the modes of applying data to the data warehouse fit into the three types of loads?
- *Initial Load*
 - Instead of loading whole data warehouse in single run, split the load into separate sub loads and run each of these sub loads as single loads. (every load run creates the database tables from scratch)
 - In this way, we will be using **load** mode.
 - For the first run of the initial load of a particular table, use the **load** mode.
 - All further runs will apply the incoming data using the **append** mode.
 - Drop the indexes prior to the loads to make the loads go quicker.
 - Rebuild or regenerate the indexes when the loads are complete.

Data Loading

- How the modes of applying data to the data warehouse fit into the three types of loads?
- *Incremental Load*
 - These are the applications of ongoing changes from the source systems tied to specific time stamps.
 - In **constructive merge** mode, if the primary key of an incoming record matches with the key of an existing record, the existing record is left in the target table as is and the incoming record is added and marked as superseding the old record.
 - If the time stamp is also part of the primary key or if the time stamp is included in the comparison between the incoming and the existing records, then **constructive merge** may be used to preserve the periodic nature of the changes.
 - The change to a dimension table record is **Type 1 change** i.e. **to correct an error** in the existing record. The existing record must be replaced by the corrected incoming record, so you may use the **destructive merge** mode.

Data Loading

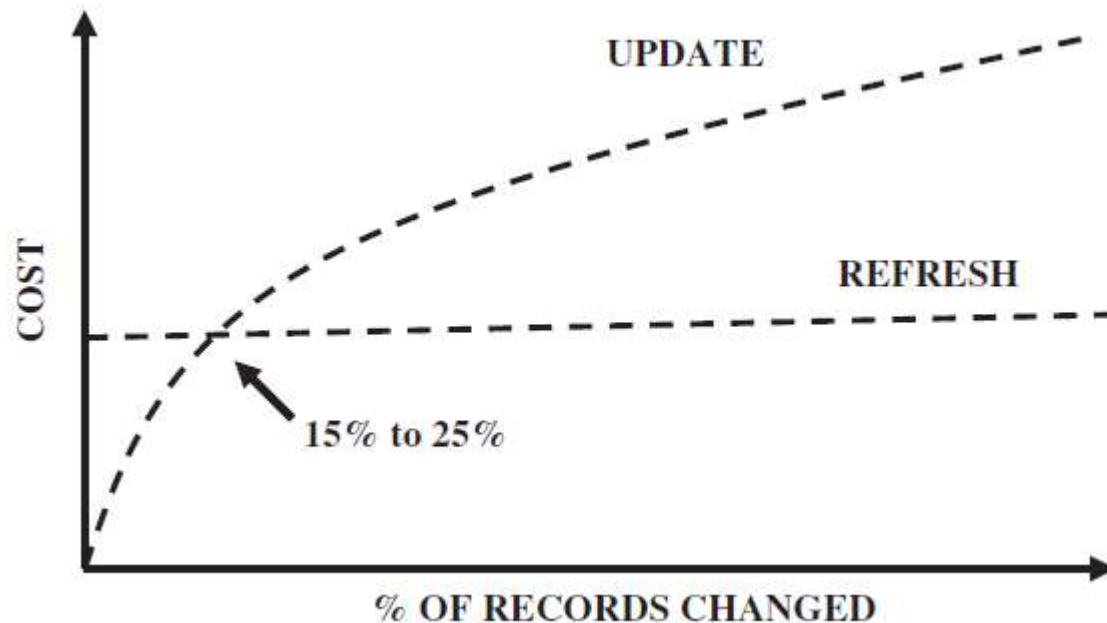
- How the modes of applying data to the data warehouse fit into the three types of loads?
- ***Full Refresh***
 - Involves periodically rewriting the entire data warehouse.
 - You may also do partial refreshes to rewrite only specific tables.
 - Full refresh is similar to the initial load.
 - Data exists in the target tables before incoming data is applied. The existing data must be erased before applying the incoming data.
 - The **load** and **append** modes are applicable to full refresh.

Data Loading

- After the initial load, you may maintain the data warehouse and keep it up to date by using two methods:
 - **Refresh**—complete reload at specified intervals.
 - **Update**—application of incremental changes in the data sources.
 - Refresh is a much simpler option than update
 - Refresh involves the periodic replacement of complete data warehouse tables.
 - Refresh jobs can take a long time to run.
 - To use the update option,
 - You have to devise the proper strategy to extract the changes from each data source
 - Then you have to determine the best strategy to apply the changes to the data warehouse.

Data Loading

- The **cost of refresh remains constant** irrespective of the number of changes in the source systems.
 - If the number of changes increases, the time and effort for doing a full refresh remain the same.
- The **cost of update varies** with the number of records to be updated.



After the initial load, the data warehouse is kept up-to-date by

REFRESH - complete reload at specified intervals

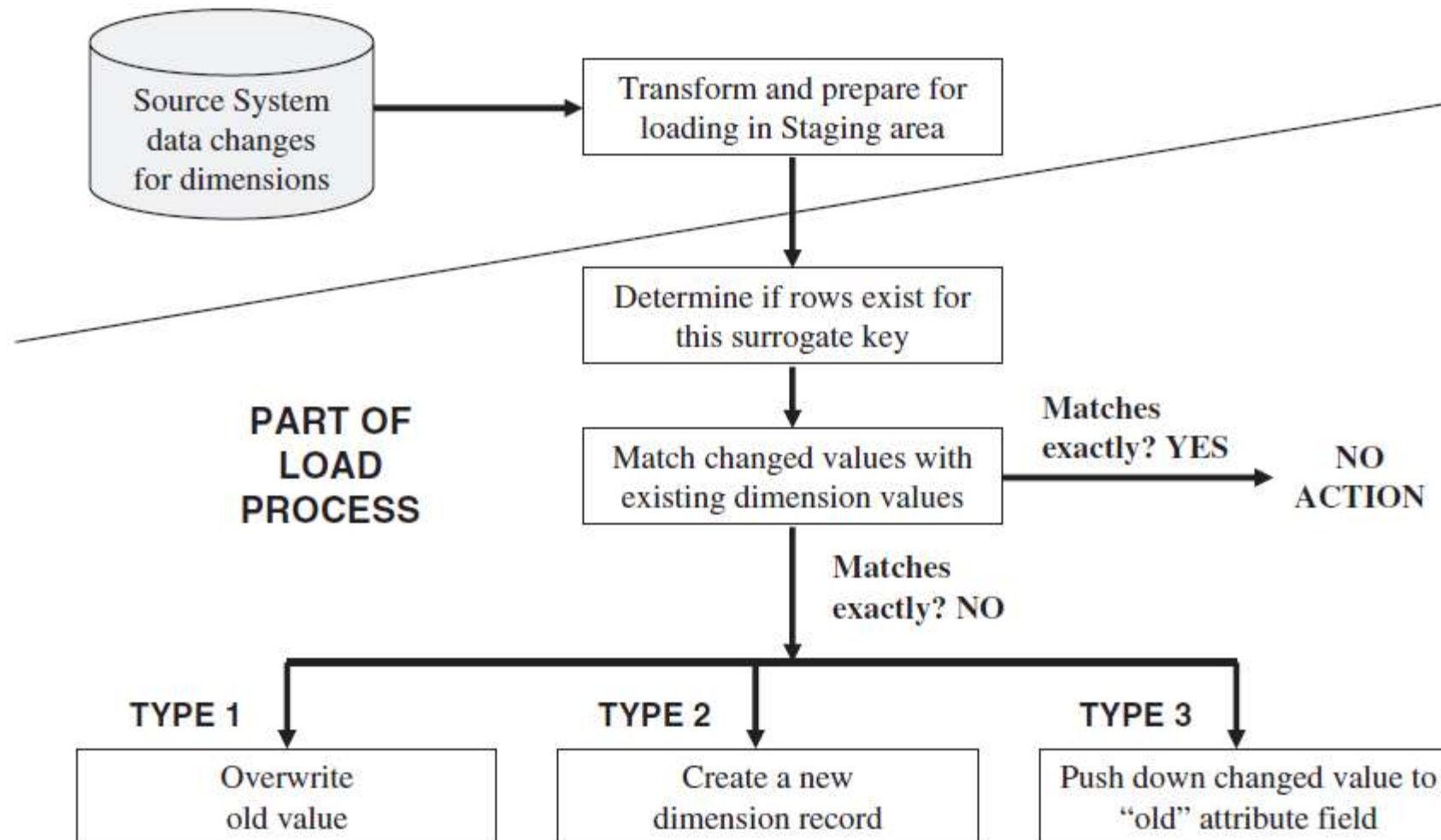
UPDATE - application of incremental changes

Data Loading

- ***Procedure for Loading Dimension Tables***

- The procedure for maintaining the dimension tables includes two functions:
 1. The initial loading of the tables;
 2. Applying the changes on an ongoing basis.
- There are two issues:
 1. About the keys of the records in the source systems
 2. The keys of the records in the data warehouse.
- In the data warehouse, you use system generated keys.

Data Loading



Data Loading

- ***Procedure for Loading Fact Tables***

- The key of the fact table is the concatenation of the keys of the dimension tables.
- Therefore, dimension records are loaded first. Then, before loading each fact table record, you have to create the concatenated key for the fact table record from the keys of the corresponding dimension records.

- **History Load of Fact Tables:**

- Identify historical data useful and interesting for the data warehouse.
- Define and refine extract business rules.
- Capture audit statistics to tie back to the operational systems.
- Perform fact table surrogate key look-up.
- Improve fact table content.
- Restructure the data.
- Prepare the load files.

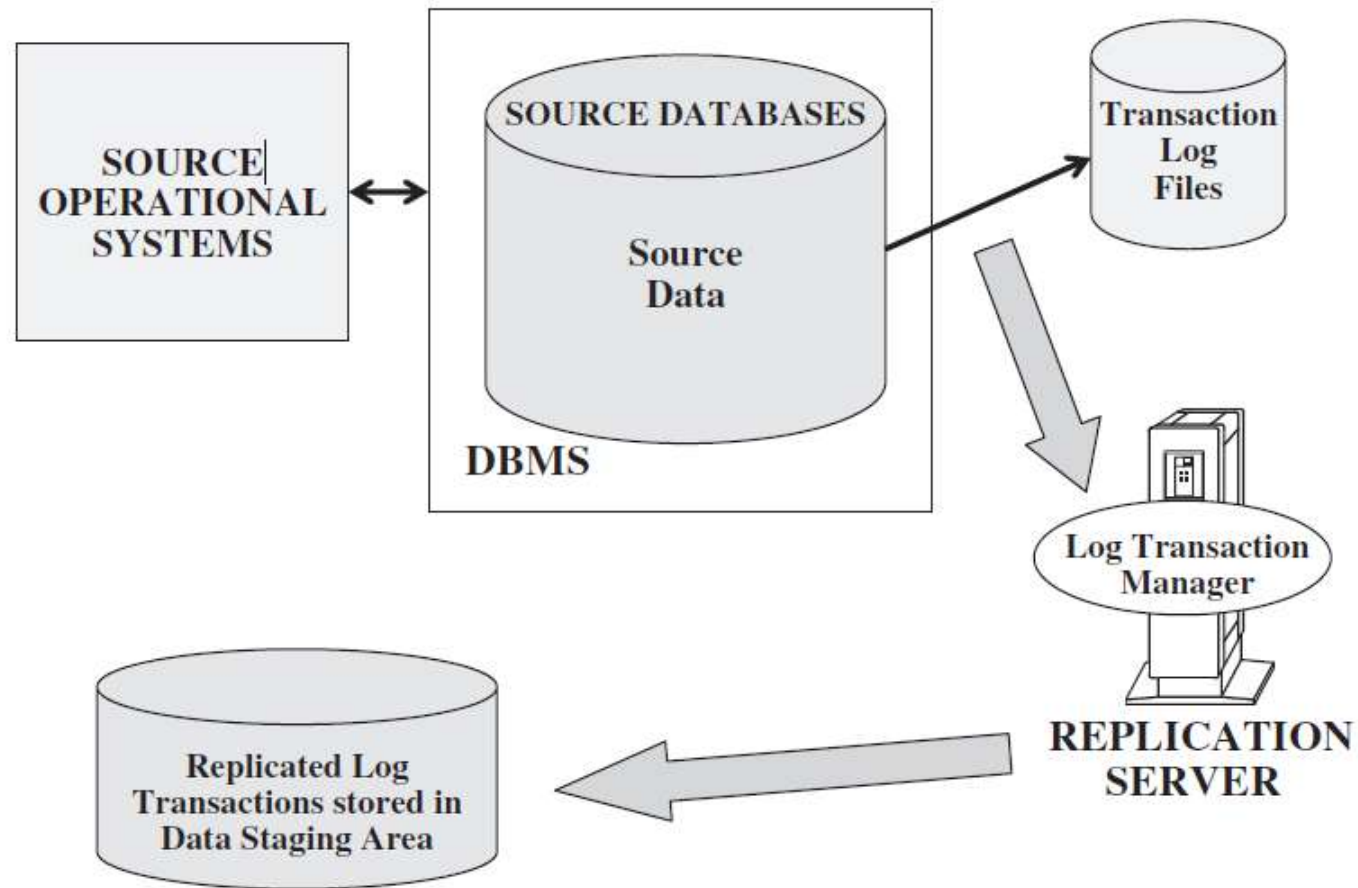
Data Loading

- *Procedure for Loading Fact Tables*
 - **Incremental Load for Fact Tables:**
 - Incremental extracts for fact tables
 - Consist of new transactions
 - Consist of update transactions
 - Use database transaction logs for data capture
 - Incremental loads for fact tables
 - Load as frequently as feasible
 - Use partitioned files and indexes
 - Apply parallel processing techniques

Data Replication

- Data replication is simply a method for creating copies of data in a distributed environment.
- The broad steps for using replication to capture changes to source data:
 - Identify the source system database table
 - Identify and define target files in the staging area
 - Create mapping between the source table and target files
 - Define the replication mode
 - Schedule the replication process

Data Replication



ETL Tools

- ETL tools fall into the following three broad functional categories:

1. Data transformation engines

- Captures data from a designated set of source systems at user-defined intervals, performs elaborate data transformations, sends the results to a target environment, and applies the data to target files.

2. Data capture through replication

- The changes to the source systems captured in the transaction logs are replicated in near real time to the data staging area for further processing.
- Some tools provide the ability to replicate data through the use of database triggers.

3. Code generators

- These tools directly deal with the extraction, transformation, and loading of data.
- Enable the process by generating program code to perform these functions.